Unit: 6 Unit 9: Data Statistical Analysis
Topic: Box Whisker (Dispersion)

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. What is the main purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the median and quartiles of a dataset
(C) To visualize the mode of a dataset
(D) To compare two datasets
(E) To calculate the standard deviation

Q2. Which part of a box-and-whisker plot represents the interquartile range?
(A) The box
(B) The whiskers
(C) The median line
(D) The outliers
(E) The box and whiskers

Q3. What is the Q_1 value in a box-and-whisker plot?
(A) The median of the dataset
(B) The lower quartile of the dataset
(C) The upper quartile of the dataset
(D) The mode of the dataset
(E) The range of the dataset

Q4. How do you calculate the IQR?
(A) $IQR = Q_3 - Q_1$
(B) $IQR = Q_1 - Q_3$
(C) $IQR = Q_3 + Q_1$
(D) $IQR = \frac{Q_3+Q_1}{2}$
(E) $IQR = \frac{Q_3-Q_1}{2}$

Q5. What is an outlier in a box-and-whisker plot?
(A) A data point within $1.5 \cdot IQR$ of Q_1 or Q_3
(B) A data point beyond $1.5 \cdot IQR$ from Q_1 or Q_3
(C) A data point equal to Q_1 or Q_3
(D) A data point between Q_1 and Q_3
(E) A data point that is the median

Q6. What does the whisker represent in a box-and-whisker plot?
(A) The range of the dataset
(B) The interquartile range
(C) The median and quartiles
(D) The minimum and maximum values, excluding outliers
(E) The mode of the dataset

Q7. Why is it important to identify outliers when analyzing data?
(A) Because they are the most common data points
(B) Because they can affect the mean of the dataset
(C) Because they are always errors in measurement
(D) Because they can indicate unusual patterns or trends
(E) Because they are irrelevant to the analysis

Q8. What is the Q_3 value in a box-and-whisker plot?
(A) The lower quartile of the dataset
(B) The median of the dataset
(C) The upper quartile of the dataset
(D) The mode of the dataset
(E) The range of the dataset

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the dataset: $2, 4, 6, 8, 10, 12, 14, 16, 18, 20$. Identify Q_1, Q_3, and the IQR. Provide your plot and calculations.

Q10. Explain how the interquartile range (IQR) is used to detect outliers in a dataset. Use the formula $IQR = Q_3 - Q_1$ and provide an example.

Chef's Tips

- Tip 1: Ensure students understand the concept of quartiles and the interquartile range.
- Tip 2: Remind students to carefully calculate the IQR and identify outliers based on the $1.5 \cdot IQR$ rule.
- Tip 3: Encourage students to practice constructing box-and-whisker plots with various datasets to reinforce their understanding.